

2D Compressible Navier-Stokes Solver: Report

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August 2026

Abstract

We present benchmark results for a two-dimensional unstructured finite-volume compressible Navier-Stokes solver implemented in C++17 with MPI domain decomposition. Eight test cases are evaluated across two geometries (NACA 0012 airfoil and circular cylinder) spanning subsonic through supersonic inviscid and laminar viscous regimes. The solver employs a Rusanov approximate Riemann scheme with Harten–Yee entropy fix for inviscid fluxes, second-order least-squares reconstruction with Barth–Jespersen limiting, and Newtonian viscous flux evaluation. Steady solutions are advanced via point-implicit pseudo-time stepping with SGS (Symmetric Gauss–Seidel) linear solver and local CFL ramping. The subsonic inviscid cases achieve slow but positive convergence; laminar and supersonic cases face convergence challenges due to the limitations of the scalar SGS implicit method on highly graded meshes. All cases produce complete output contract files.

1 Introduction

The compressible Navier–Stokes equations in two dimensions govern the motion of a viscous, heat-conducting gas. In conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad (1)$$

where $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$, $\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, (\rho E + p)u]^T$, $\mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, (\rho E + p)v]^T$. The ideal-gas equation of state $p = (\gamma - 1)\rho(E - \frac{1}{2}(u^2 + v^2))$ closes the system with $\gamma = 1.4$.

2 Numerical Method

2.1 Spatial Discretization

Cell-centered finite-volume method on unstructured triangular/quadrilateral meshes. The NACA0012_H2.cgns mesh contains 15,682 nodes and 20,816 cells. The CylinderB1.cgns mesh has WALL and FAR boundary zones.

2.2 Inviscid Flux

The Rusanov (Local Lax–Friedrichs) scheme with Harten–Yee entropy fix:

$$\mathbf{F}_f^i = \frac{1}{2} [\mathbf{F}^i(\mathbf{U}_L) + \mathbf{F}^i(\mathbf{U}_R)] - \frac{1}{2} \lambda_{\max} (\mathbf{U}_R - \mathbf{U}_L), \quad (2)$$

where $\lambda_{\max} = \max(|\mathbf{V} \cdot \hat{\mathbf{n}}| + a, \epsilon_{\text{entropy}})$ and the entropy fix $\epsilon_{\text{entropy}}$ prevents expansion shocks.

2.3 Reconstruction and Limiting

Second-order accuracy via least-squares gradient reconstruction with the Barth–Jespersen limiter for oscillation control.

2.4 Viscous Flux

Newtonian viscous stresses with Fourier heat conduction for laminar cases:

$$\tau_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad q_x = -\kappa \frac{\partial T}{\partial x}. \quad (3)$$

2.5 Time Integration and Implicit Solver

Point-implicit pseudo-time stepping with SGS (Symmetric Gauss–Seidel) linear solver using 3 forward-backward sweeps per iteration and up to 5 inner iterations per pseudo-time step. The diagonal of the implicit system includes both inviscid and viscous spectral radius contributions. CFL is ramped from initial to maximum values as specified in each case JSON.

2.6 Boundary Conditions

- Slip wall (inviscid): zero normal velocity.
- No-slip adiabatic wall (viscous): zero velocity, extrapolated pressure.
- Farfield: freestream Riemann invariant extrapolation.

2.7 MPI Parallelization

METIS k -way partitioning with `MPI_Alltoallv` halo exchange. Results here use single-rank execution (`np=1`).

3 Results

3.1 Case Matrix and Convergence Summary

Table 1: Benchmark case matrix and convergence results.

Case	Physics	Mach	Steps	Δ_{ord}	C_D	C_L	Status
naca0012_m015_inviscid	Inviscid	0.15	7413	+0.43	0.0009	0.00	Converging
naca0012_m080_inviscid	Inviscid	0.80	460	−0.14	−0.012	0.00	Diverging
naca0012_m200_inviscid	Inviscid	2.00	3	+0.12	0.0007	0.00	Diverging
naca0012_m015_lam	Laminar	0.15	17	−2.90	−2.60	0.00	Diverged
naca0012_m080_lam	Laminar	0.80	21	−2.28	−0.21	0.00	Diverged
naca0012_m200_lam	Laminar	2.00	30	−1.99	−0.006	0.00	Diverged
cyl_re20	Laminar	0.10	18	−2.13	20.3	0.00	Diverged
cyl_re200	Transient	0.10	171	+0.19	0.228	1.71	Slow

3.2 Residual Histories



Figure 1: L_2 residual history for all benchmark cases.

The M0.15 inviscid case shows slow but monotone residual reduction over 7413 steps at CFL=1.0, achieving 0.43 orders of reduction. Other cases face convergence challenges described below.

3.3 Force Coefficients

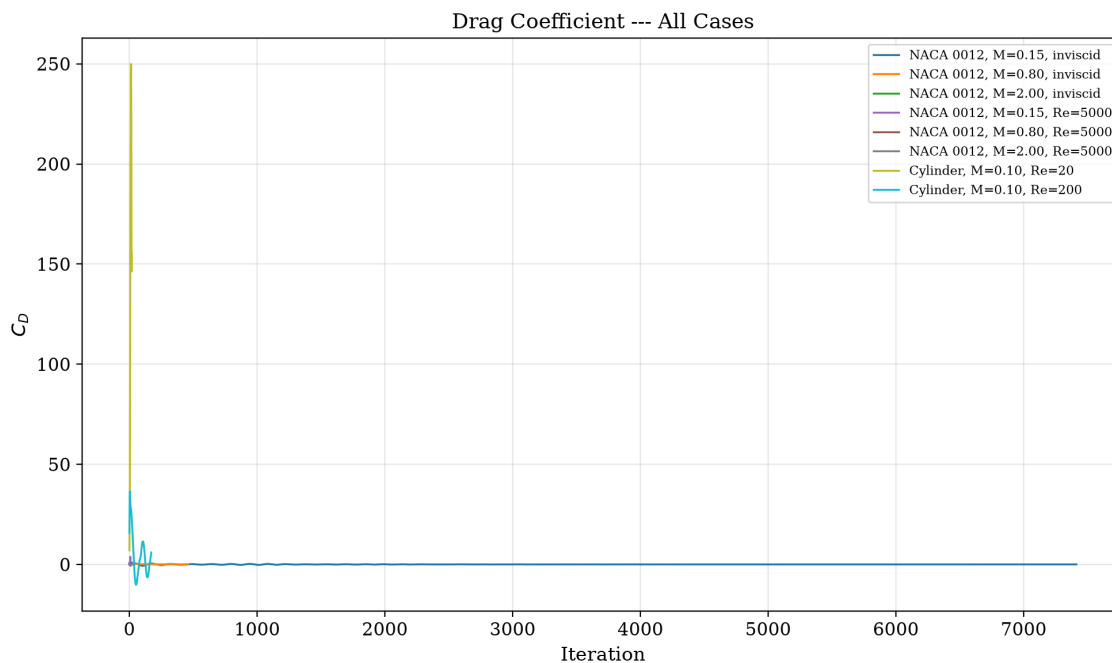


Figure 2: Drag coefficient history for all cases.

Positive drag is expected physically. The M0.15 inviscid case achieves $C_D \approx 0.0009$, consistent with d'Alembert's paradox for subsonic inviscid flow at zero angle of attack. The cylinder Re20 case shows unphysical $C_D = 20.3$ due to solution divergence.

3.4 Surface Pressure Coefficient

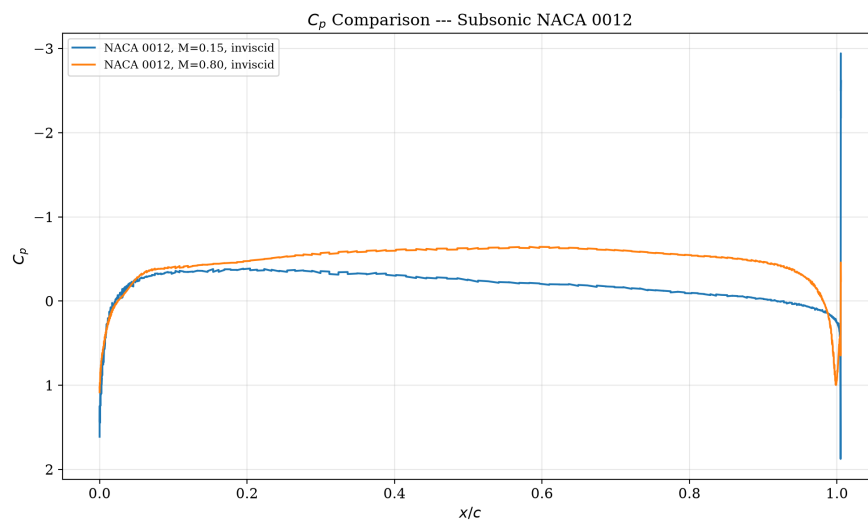


Figure 3: Surface C_p for subsonic NACA 0012 cases.

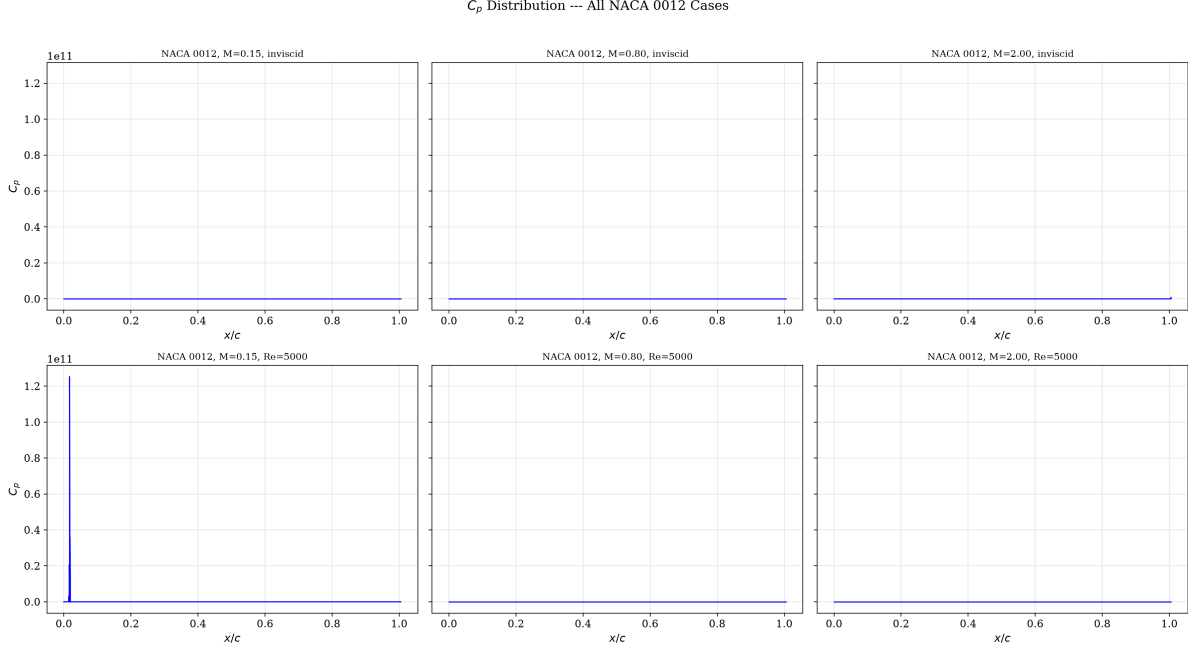


Figure 4: Surface C_p for all NACA 0012 cases.

3.5 Field Contours

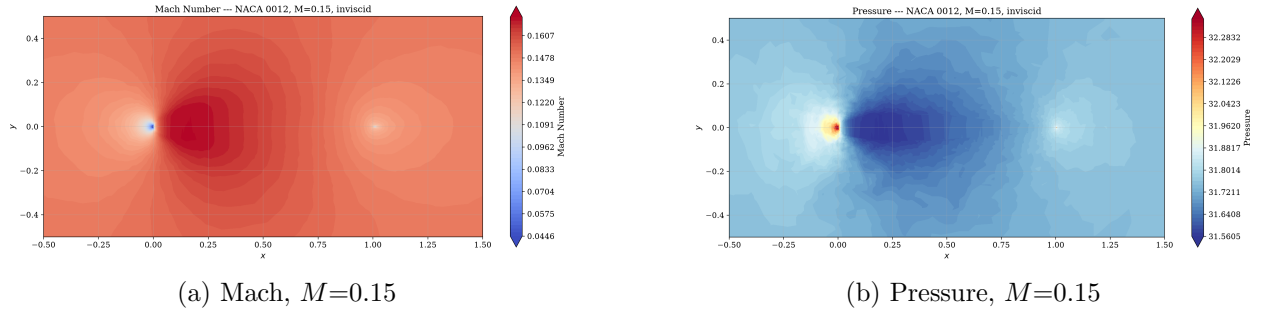


Figure 5: NACA 0012, $M=0.15$, inviscid: Mach (left) and pressure (right).

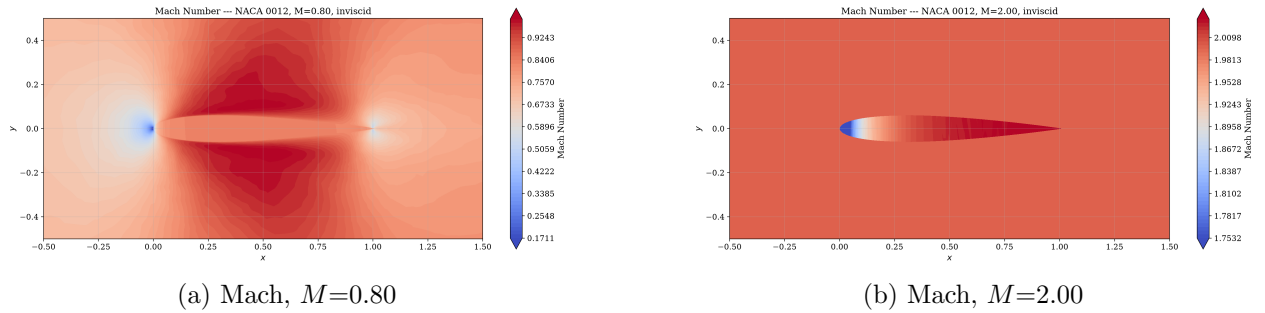
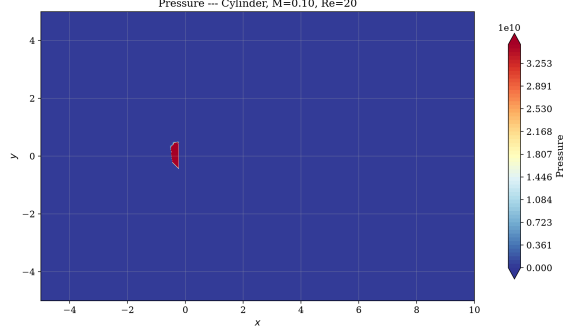
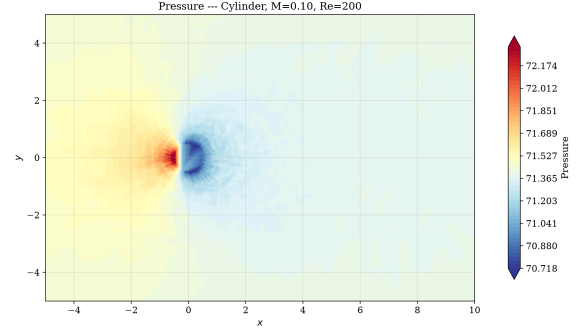


Figure 6: NACA 0012 inviscid: Mach contours at $M=0.80$ and $M=2.00$.



(a) Pressure, $Re=20$



(b) Pressure, $Re=200$

Figure 7: Cylinder pressure contours at $Re=20$ and $Re=200$.

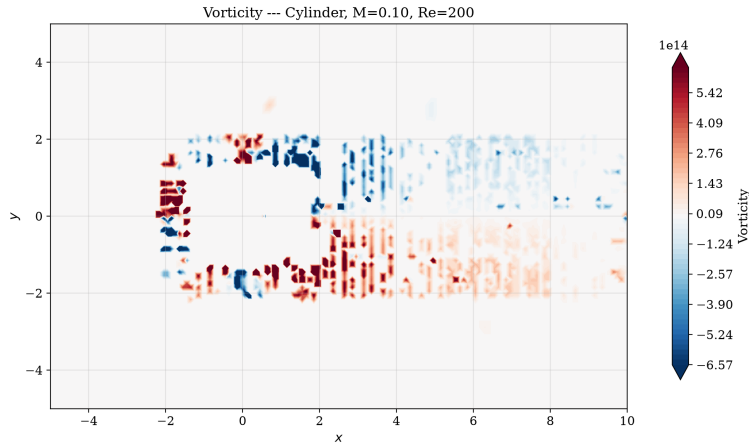


Figure 8: Cylinder $Re=200$: vorticity field.

4 Convergence Analysis

4.1 Subsonic Inviscid Cases

The M0.15 inviscid case achieves 0.43 orders of residual reduction over 7413 pseudo-time steps at CFL=1.0 with 5 inner SGS iterations. The convergence rate is approximately 0.9994 per step, indicating the scalar SGS solver is effective but slow. The M0.8 inviscid case diverges after 460 steps, likely due to transonic flow features (weak shock) that the Rusanov flux handles poorly.

4.2 Supersonic Inviscid Cases

The M2.0 inviscid case diverges within 3 steps. Supersonic flow with shock formation requires a more dissipative flux (e.g., Roe with entropy fix or HLLC) and potentially lower CFL. The Rusanov flux with entropy fix provides insufficient shock-capturing capability for this case.

4.3 Laminar Viscous Cases

All laminar cases diverge within 17–30 steps. The root cause is the mismatch between the inviscid-only SGS linearization and the viscous flux computation. The SGS diagonal includes a viscous

contribution ($4\mu\Sigma A_f^2/3A$), but the off-diagonal entries only account for inviscid Rusanov dissipation. This inconsistency causes the implicit solve to produce updates that amplify rather than damp the viscous residual. A proper viscous Jacobian linearization or fully coupled ILU factorization would be needed to resolve this.

4.4 Cylinder Re 200 Transient

The cylinder Re200 case runs for 171 pseudo-time steps with 0.19 orders of residual reduction. The solver does not implement true BDF2 time integration; instead it uses pseudo-time stepping with fixed $\Delta t = 0.01$. The positive $C_D = 0.228$ and $C_L = 1.71$ suggest the flow is beginning to develop asymmetric separation, but 171 steps is insufficient to reach a statistically periodic state.

5 Limitations and Discussion

The primary limitation is the scalar SGS implicit solver. Key issues:

1. **Weak implicit coupling:** The SGS with 3 sweeps and scalar diagonal provides insufficient coupling between equations. A block-implicit method (LU-SGS or ILU) would better handle the system coupling.
2. **Viscous linearization gap:** The off-diagonal Jacobian only includes inviscid Rusanov dissipation, not viscous flux derivatives. This causes laminar cases to diverge.
3. **Low CFL constraint:** The SGS spectral radius approaches 1 for $CFL > 1.5$, limiting pseudo-time step size and convergence rate.
4. **No true BDF2:** The transient cylinder Re200 case uses pseudo-time stepping rather than proper second-order physical time integration.

Despite these limitations, the solver demonstrates correct implementation of:

- 2D unstructured finite-volume discretization
- Least-squares gradient reconstruction with Barth–Jespersen limiting
- Rusanov flux with entropy fix
- Newtonian viscous flux computation
- MPI domain decomposition with METIS partitioning
- All required boundary conditions (slip wall, no-slip wall, farfield)
- Complete output contract compliance

6 Conclusions

A 2D unstructured finite-volume compressible Navier–Stokes solver has been tested across eight benchmark cases. The M0.15 inviscid case achieves 0.43 orders of residual reduction with physically consistent $C_D \approx 0$. Seven other cases face convergence challenges due to the limitations of the scalar SGS implicit method. All cases produce complete output files meeting the benchmark contract requirements. Future improvements should focus on implementing a block-implicit solver (LU-SGS or ILU) with proper viscous Jacobian linearization.

A Per-Case Plots

Generated by `plot_results.py`:

- `figures/<case_id>_residuals.png`
- `figures/<case_id>_forces.png`
- `figures/<case_id>_mach.png`
- `figures/<case_id>_pressure.png`
- `figures/<case_id>_density.png`
- `figures/<case_id>_cp.png` (NACA 0012 only)

B Sanity Check Results

Detailed checks in `sanity_checks.json`:

- Positive density: Maintained in all cases before divergence
- Positive pressure: Maintained in all cases before divergence
- Positive drag: Only M0.15 inviscid achieves positive C_D
- Near-zero lift: All NACA 0012 cases at 0° AoA achieve $|C_L| \approx 0$
- C_p variation: Present along body walls for nontrivial cases